Investigating Privacy Transparency Gaps in Connected Vehicle Interfaces: A Think-Aloud Study Design

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Abstract: Connected vehicles collect and transmit an increasing volume of driver and passenger data, spanning location, driving behaviour, in-cabin audio, and biometric signals. Manufacturer privacy notices disclose these practices in aggregate, but it remains unclear whether drivers and passengers can understand what is collected, why it is collected, and how to exercise meaningful control through the in-vehicle infotainment system. Previous work has audited privacy policies and surveyed general concerns, but few studies have examined how users make sense of privacy information at the point of interaction. This work-in-progress reports on the design and pilot phase of a task-based think-aloud study evaluating current CV privacy interfaces. Participants attempt concrete privacy-related tasks, such as identifying what data is collected, discovering the purposes of collection, and adjusting sharing preferences. We describe the task design, interface stimuli, and preliminary observations from pilot sessions, and discuss the implications for designing privacy interfaces that better support drivers and front-seat passengers in understanding and managing CV data collection. The full study will inform a subsequent co-design workshop toward improved privacy interface designs.

Keywords: Privacy, Connected Vehicles, User Study, Interface

1. Introduction

When a driver buys a modern connected vehicle (CV), they enter into a data relationship whose terms are difficult to inspect. The car records where they drive, how they drive, what they say in the cabin, and how they interact with the infotainment system. Some of this data is legally required to be collected, most of it is not [1], [2], [3]. Some data is shared only with the manufacturer, while some flows to insurance companies and data brokers and, as recent enforcement actions have shown, has been sold to third parties without meaningful user consent [4], [5]. The interfaces through which drivers and passengers might understand and manage this data relationship, the in-vehicle infotainment display and the manufacturer's companion mobile application, are the first places most users encounter their vehicle's privacy practices. Whether these interfaces actually equip users to make sense of what is happening remains an open question.

Prior research has largely approached this question at two levels of abstraction. Regulatory and audit-focused studies have documented what manufacturers disclose in their written privacy policies [4], [6]. Survey-based research has measured general privacy concerns among CV users [7], [8]. Both bodies of work are valuable, but neither examines the point at which most users would actually encounter privacy information. The moment they open a settings menu, read a disclosure screen, or attempt to change a preference. The usable privacy literature in adjacent domains, such as mobile applications, IoT devices, and smart speakers, has repeatedly shown that this point-of-interaction level is where transparency succeeds or fails [9]. To our knowledge, no published study has systematically examined CV privacy interfaces through this lens.

To address this gap, we are pursuing a three-part research plan structured around the following research questions:

- **RQ1.1:** What transparency gaps do drivers encounter when attempting to understand and manage the data collected by their connected vehicle through in-vehicle privacy interfaces?
- **RQ1.2:** How do front-seat passengers, who interact with the same interfaces without being the account-holding data subject, perceive and respond to those same transparency gaps?
- **RQ2:** What interface design principles and features best support privacy transparency and meaningful opt-out for vehicle occupants and nearby third parties?
- **RQ3:** Does a redesigned interface that incorporates those principles improve users' perceived transparency,

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trust, and sense of control compared to a baseline system?

Each research question corresponds to one step of the plan. **RQ1.1** and **RQ1.2** are addressed by a formative task-based think-aloud study, in which drivers and front-seat passengers attempt concrete privacy-related tasks with current CV interfaces, identifying what data is collected, discovering the purposes of that collection, checking third-party sharing, and adjusting privacy settings, while narrating their reasoning. **RQ2** will be addressed through co-design workshops in which participants from the think-aloud study collaboratively develop interface concepts that respond to the transparency gaps it documents. **RQ3** will be addressed by implementing a prototype interface based on the workshop outcomes and evaluating it with the same task protocol used in the first study, enabling a direct comparison between the redesigned interface and current manufacturer baselines.

This work-in-progress paper reports on the first step of this plan (**RQ1.1** & **RQ1.2**). The design and pilot phase of the think-aloud study. We describe the study protocol, present preliminary observations from pilot sessions, and outline the refinements these observations motivate for full data collection. Our contribution is methodological and formative. A task-based approach to evaluating CV privacy interfaces that will serve as both the diagnostic instrument for **RQ1** and the evaluation instrument for **RQ3**, together with early empirical evidence of the transparency gaps that motivate our forthcoming design work.

2. Background

2.1 Toward transparent privacy design

A substantial body of research in usable privacy has established that meaningful transparency requires more than the disclosure of data collection practices, it requires design that supports users in making sense of what is being collected and acting on what they learn. Early work on privacy notices demonstrated that traditional long-form privacy policies fail on both counts: users do not read them, and cannot accurately answer questions about their content when they do [10], [11]. In response, researchers have proposed and evaluated a range of alternative disclosure formats and control mechanisms.

The privacy nutrition label [12], [13] introduced the idea that standardised, visual disclosures could allow users to compare privacy practices across products in the same way nutrition labels support comparison across foods. Follow-up work extended this approach to IoT devices [14] and directly informed the industry deployment of Apple's App Store privacy labels and Google's Data Safety section, though subsequent audits have shown that self-reported labels frequently diverge from actual data practices [15].

Parallel to the label tradition, work on contextual and purpose-specific consent has demonstrated that users' privacy preferences depend heavily on the purpose for which data is collected, not merely the data category [16], [17]. Systems that surface purpose at the point of interaction, through just-in-time prompts, personalised assistants, or periodic access-count nudges, have been shown to shift both user comprehension and user behaviour [9], [18]. Across these works, a consistent finding emerges: privacy interfaces that are simple, comparable, and grounded in specific data-use contexts outperform interfaces that provide the same information in more abstract or aggregated forms.

The success of these approaches in mobile and IoT contexts raises the question of whether analogous designs could improve privacy transparency in CVs. Vehicles share many of the technical properties that motivated this research, such as continuous sensing, multi-party data flows, opaque back-end processing, and consent frameworks that in practice reduce to all-or-nothing acceptance. Yet the design patterns validated in the usable privacy literature have not been systematically applied or evaluated in the vehicle domain.

2.2 Privacy in connected vehicles

Recent work has documented the scale and opacity of data collection in CVs. Independent reviews of manufacturer privacy practices have shown that CVs collect location, driving behaviour, in-cabin audio, biometric signals, and infotainment interactions, and that manufacturer disclosures around these practices are typically incomplete, difficult to compare across vendors, and structured around all-or-nothing consent [4]. Regulatory action has reinforced these findings: the California Attorney General's 2026 settlement with General Motors concerning the sale of OnStar driving data to insurance data brokers demonstrated that stated privacy practices and actual data flows can diverge substantially [5].

User-facing research in this area has focused primarily on measuring general privacy concerns and their relationship to service adoption. Walter and Abendroth [7] extended the Technology Acceptance Model with privacy constructs and found that privacy-related factors were more significant predictors of connected-service adoption than perceived usefulness or ease of use. Subsequent survey and experimental work have examined trade-offs between privacy and convenience [8], willingness to pay for privacy protections in automated vehicles [19], and multi-stakeholder perceptions in in-cabin sensing scenarios [20]. Our own prior work has audited manufacturer privacy policies across manufacturers and jurisdictions and measured cross-cultural comfort with CV data collection [*1]. We found that users are more concerned with CV privacy the further from control they are placed. These studies collectively establish that CV privacy is a substantive and contested area.

However, what has not yet been examined is the specific point at which most drivers and passengers would encounter privacy information, the interface. Prior studies have documented what manufacturers claim about data practices and what users report feeling about those practices, but have largely not examined what users do when they attempt to inspect or manage privacy through vehicle interfaces.

[*1] This work is an extension of CSS 2025's [21] and VehicleSec 25's [6].

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2.3 Automotive interface research

A small but growing body of work has begun to treat privacy as a design object within vehicle interfaces, rather than solely as a policy or acceptance factor. Ihme et al. [22] developed and evaluated a data privacy concept for a frustration-aware in-vehicle system, addressing how affect-sensing vehicles should communicate data practices to users. Kim et al. [23] proposed a context-aware "private call" mode for in-vehicle voice assistants, designing around the concern that always-on microphones capture conversations users consider private. Most directly related to transparency, Sandhaus et al. [24] showed in an online study (N = 42) that a transparent step-by-step on-boarding design for camera-based in-car health monitoring significantly reduced perceived creepiness and increased trust in data use compared with traditional terms-and-conditions text and nudge-based designs, evidence that transparent disclosure design measurably shifts user perceptions in the vehicle context. In parallel, interview-based work by Dowthwaite et al. [25] derived design recommendations for legibility and agency in car data interfaces, concluding that data-sharing settings should be granular, continuously accessible, and clearly signaled in both the companion app and the dashboard.

These studies establish that privacy-transparent design is both feasible and consequential in vehicles. However, they share two limitations that motivate the present study. First, each addresses a single feature or hypothetical system, an affect-aware assistant, a health-monitoring camera, a voice-call mode, typically evaluated through online prototypes or interviews rather than through interaction with the privacy interfaces manufacturers actually deploy. Second, their outcome measures center on acceptance and trust rather than on whether users can successfully find, understand, and act on privacy information. The task-level questions that the mobile and IoT usable-privacy literature has shown to be where transparency succeeds or fails [9]. Task-based evaluation of deployed CV privacy interfaces, of the kind standard in adjacent domains, remains absent. This positions the present study, to characterise, through concrete tasks with real interfaces, how drivers and front-seat passengers make sense of the privacy information and controls currently available to them.

3. Methods

We designed a task-based think-aloud study to investigate how drivers and front-seat passengers make sense of the privacy interfaces of current connected vehicles. The study consists of a single session combined with a pre-study questionnaire, a task-based think-aloud interaction with real CV interfaces, and a semi-structured follow-up interview. This paper reports on the study design and initial observations from N = 2 pilot sessions conducted during August 2026.

3.1 Study Design

We use a within-subjects design in which each participant attempts a fixed set of tasks. Each session lasts approximately 30 minutes and consists of three phases: (1) pre-study questionnaire, (2) task-based think-aloud interaction, and (3) semi-structured follow-up interview. Additionally, participants are asked if they would be interested in a follow up workshop session. Sessions are audio-recorded and notes are recorded by the interviewer.

3.1.1 Pre-study questionnaire

The pre-study questionnaire captures participants background and prior engagement with CVs. These are conducted online, using Google Forms, prior to the in person task based study. The questionnaire consists of six short sections covering:

(1) Informed consent and participant requirements.
(2) Demographics (age, gender, education, country of residence, country of primary vehicle experience).
(3) Familiarity with Connected Vehicles (familiarity with the term and features).
(4) Vehicle usage (frequency as driver and as front-seat passenger, ownership status, and vehicle features).
(5) IUIPC-8 [26] scale questions for general privacy attitudes.
(6) Scheduling times and logistical information (when the participants are available to conduct the task based study).

The pre-study questionnaire takes approximately 5-7 minutes to complete.

3.1.2 Task-based think-aloud interaction

Participants complete five to six, based on their main use of CVs, privacy-related tasks with the study's interface. Each task is explained aloud by the researcher to the participant. Participants are instructed to think aloud continuously as they attempt each task, following standard think-aloud protocol conventions [27]. A short warm-up task (counting windows in the participant's childhood home while narrating their reasoning) precedes the study tasks to calibrate participants to the think-aloud method without privacy-related priming.

The six tasks target distinct dimensions of privacy transparency:

(1) **Data disclosure.** Using the in-vehicle screen, identify what types of information the vehicle collects about the driver and their driving.
(2) **Purpose identification.** Determine why the vehicle collects location data specifically, and what the location data is used for.
(3) **Third-party sharing.** Determine whether the vehicle shares data with any entities other than the manufacturer.
(4) **Privacy configuration.** Configure the vehicle's settings such that only data strictly required for safe vehicle operation is collected.
(5) **Passenger data collection.** From the perspective of a front-seat passenger, determine what data the vehicle collects about the passenger specifically (e.g. from

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cabin microphone or driver-facing camera)[*2].

(6) **Voice data handling.** Determine whether the vehicle records what is said in the cabin and what happens to any such recording.

Tasks are capped at approximately 4 minutes each. Participants who are unable to make further progress within the cap are moved to the next task. The researcher uses only non-leading probes during tasks (e.g., "What are you looking for?", "What did you expect to happen?"). Task success is coded by the researcher on a four-point scale distinguishing complete success, partial success (goal reached but with clarifying prompts or misinterpretation), failure (participant gave up or exceeded the time cap), and interface impossibility (participant correctly identified that the task cannot be completed with the available interface).

3.1.3 Semi-structured follow-up interview

After the task block, participants complete a semi-structured interview lasting approximately 10–15 minutes. The interview covers four blocks:

(1) Immediate reactions to specific tasks (which felt hardest and why, what participants found surprising).

(2) Interface-level reflection (comparative evaluation across the interfaces used, information participants sought but could not find).

(3) Concern elicitation (privacy-related concerns arising from the tasks, and whether these concerns differ from the participant's pre-session views).

(4) Design projection and Study 2 recruitment (invitation to participate in a follow-up co-design workshop).

For passenger participants, an additional short block probes the driver–passenger asymmetry surfaced by Task 5. Interview probing follows established qualitative interviewing practice, with silence and open follow-ups preferred over evaluative or leading questions.

3.2 Participant recruitment and eligibility

Participants for the pilot study were recruited through friends and family and screened against the following eligibility criteria:

(1) Aged 18 or older.

(2) Has driven or ridden as a front-seat passenger in a connected vehicle (defined as a vehicle manufactured in 2015 or later with manufacturer-provided companion app or in-vehicle infotainment with data connectivity) within the past 12 months.

(3) Sufficient fluency in English/Japanese.

(4) Consented to participation in the study.

We target a final sample of N = 2, drivers(1) and front-seat passengers(1). For the pilot study, participants did not receive compensation for their time and were informed of this before participation in the study.



Fig. 1: 2015 Audi Q5 in vehicle interface menus (Telephone, vehicle settings, navigation, and radio).



Fig. 2: 2015 Audi Q5 in vehicle control system.

3.3 Equipment and interface

Sessions are conducted in person, with the participant seated in the front passenger seat of the study vehicle, a 2015 Audi Q5. The in-vehicle interface is presented on a screen in the central console (Figure 1), which is visible and accessible from both the driver and front passenger seats. Participants interact with the interface through physical controls located on the steering wheel and in the gear-shift area (Figure 2). A central rotary knob controls selection within menus, and four buttons surrounding the knob select menus corresponding to their on-screen orientation. Five dedicated buttons (navigation, telephone, radio, media, and the central menu) provide direct access to major functions, and a back button returns to the previous screen. The vehicle offers connected

[*2] Only participants who are primarily front seat passengers are assigned this task.

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features including Bluetooth device pairing and a SiriusXM satellite radio subscription service.

The task-based study and follow-up interviews are audio-recorded using a recording device placed inside the vehicle. A researcher observation sheet is used to note task hesitations, expressions of confusion, and moments of surprise not captured by the recordings.

3.4 Data handling and analysis

All sessions were conducted with informed consent, participants were briefed on recording, data handling, and their right to withdraw at any time. Audio recordings are transcribed and anonymised, with participants assigned code numbers and identifying details replaced with generic descriptors. Quotations presented in this paper are lightly edited for readability, removing filler words while preserving meaning. Think-aloud and interview data are analysed using reflexive thematic analysis [28], and task outcomes are coded on the four-level scheme described above and reported descriptively. For the full study, coding will be performed by two researchers with disagreements resolved through discussion, following current qualitative reporting guidance [29].

3.5 Data reported in this work-in-progress paper

This paper reports on the study design in full, together with initial observations from $N = 2$ pilot sessions completed to date. The pilot sessions are used to refine the task wording, session timing, and interview probes ahead of full data collection. Findings drawn from pilot data are presented as preliminary observations and are not intended to support inferential claims.

4. Results

We report preliminary observations from two pilot sessions (P1, P2) conducted to refine the study protocol ahead of full data collection. Given the pilot sample size, these results are descriptive and are intended to illustrate the study's analytical yield and to inform protocol refinements rather than to support generalisable claims.

4.1 Participant characteristics

Table 1 summarises the pilot participants' demographics, background knowledge, vehicle usage, and IUIPC-8 scores. Both participants were male, aged 18–29, and learned to drive in Canada. Neither owned a vehicle and both regularly used a car owned by someone else. Participants both reported at least weekly experience as driver and as front-seat passenger. Self-reported familiarity with the term "connected vehicle" was slight (P1) to moderate (P2), and both participants scored at the "basic" level on the objective knowledge check, indicating awareness that vehicles collect some data but limited knowledge of specific practices.

Both participants recorded high IUIPC-8 sub-scale scores (Collection: 4.5 and 4.75; Control: 5 and 4.5; Awareness: 5 and 5, on a 5-point scale), indicating strong general privacy concern despite limited connected-vehicle knowledge.

4.2 Task outcomes

Table 2 reports task outcomes per participant, coded on the four-level scheme described in Section 3.1.3 (complete success, partial success, failure, interface impossibility).

Neither participant achieved complete success on any of the six tasks. Tasks 1 (data disclosure) and 2 (purpose identification) yielded partial success for both participants. Participants located general settings menus and some data-related toggles but could not assemble a complete account of what the vehicle collects. Additionally, both participants were able to imply that the use of the navigation system collects location information but were unable to find any disclosure information.

Tasks 3 (third-party sharing) and 4 (privacy configuration) were coded as interface impossibility for both participants. The in-vehicle interface provided no disclosure of third-party data sharing and no controls for restricting collection to safety-required data. Both participants were able to assume this after navigating all menus and expending the task limit. We note that interface impossibility is a failure of the interface rather than of the participant. These outcomes constitute positive findings about the state of current disclosures and privacy gaps.

Task 5 (passenger data collection) was coded as interface impossibility for P1 and was not administered to P2, as they considered themselves to be primarily a driver. Task 6 (voice data handling) produced divergent outcomes. P1 was unable to locate any information about in-cabin audio recording or retention within the time cap (failure), while P2 achieved partial success as they discovered voice activated search features and implied the use of voice recognition.

In total, across the eleven administered task instances, we observed four partial successes, six interface impossibilities, one failure, and no complete successes. The predominance of interface-impossibility outcomes over participant failures suggests, preliminarily, that the transparency gaps in the pilot vehicle's interface stem primarily from absent disclosures rather than from discoverability problems with information that exists. A distinction with direct design implications, since the former cannot be remedied by improved menu structure alone.

4.3 Preliminary observations from think-aloud data

Absent disclosure is interpreted as absent collection. The most consequential observation from the pilot concerns how participants resolved missing information. Unable to find any statement about third-party data sharing (Task 3), [P1] concluded,

> "I didn't see that, so I'm assuming... they can't do that."

The same inference recurred for passenger data (Task 5),

> "I would assume that, as a passenger, it doesn't have any of my data."

In both cases, the absence of disclosure was read as evidence of absence of practice, an inference that, given documented

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Table 1: Demographics, background knowledge, driving and passenger frequency, and IUIPC 8 scores of participants.

	Age	Gender	Education	Driving location	CV Term Familiarity	CV Knowledge	Driving Frequency	Passenger Frequency	Vehicle Ownership	Col	Ctrl	Awa
P1	18-29	Male	High School	Canada	Slightly familiar	Basic	Once a week	Once a week	No, but I regularly use a car owned by someone else	4.5	5	5
P2	18-29	Male	Undergraduate degree	Canada	Moderately familiar	Basic	Several times a week	Several times a week	No, but I regularly use a car owned by someone else	4.75	4.5	5

Table 2: Coded responses to each participants task completion code.

	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6
P1	Partial success	Partial success	Interface impossibility	Interface impossibility	Interface impossibility	Failure
P2	Partial success	Partial success	Interface impossibility	Interface impossibility	N/A	Partial success

industry practices [4], [5], may leave users with materially inaccurate mental models.

The passenger role suppresses privacy action. When framed as a passenger, [P1] described disengaging from privacy management entirely,

> "If I'm a passenger in the vehicle, I'm not doing anything... I'm not messing with settings. I wouldn't go to the driver and be like — are you tracking?"

Pointing to a social barrier, beyond the interface barrier, to passenger privacy action. Notably, the task prompt itself shifted the participant's awareness,

> "Now that you're asking... it's like, what data are you collecting on passengers? Hold on a moment."

The concern was generated by the study prompt rather than pre-existing, consistent with the interface providing no cue that passenger data collection is a question worth asking.

Protective habits. Participants also brought protective habits to the session, although the interface offered little for those habits to act on. [P2] described a default strategy of disabling any setting not required for vehicle function,

> "...anything where it doesn't improve the performance of the vehicle or [is] not necessary, I default turn those off... that's what I do with all my devices."

and applied it during the session, for example disabling voice-recognition speech training on the reasoning that:

> they usually train models.

The participant also correctly distinguished the interface's data-encryption option from a data-collection control.

> "...that just means they can still do whatever they want with the data... it just means that some random person can't pull it out."

These behaviours indicate a privacy-literate user actively seeking control. The interface-impossibility outcomes on Tasks 3–5 (Table 2) reflect the interface's failure to provide corresponding controls rather than a lack of participant motivation or capability.

5. Discussion

The pilot sessions served their primary purpose: they surfaced concrete refinements to the protocol and generated candidate themes that the full study is now positioned to test. We discuss both, together with the limitations that shape how the pilot observations should be read.

5.1 Implications for users

The most consequential candidate theme from the pilot is that participants interpreted the absence of privacy disclosures as evidence of the absence of data practices. For example, participants concluded that the vehicle could not share data with third parties because no such setting existed. If this pattern replicates in the full study, it reframes the transparency problem. Incomplete vehicle privacy interfaces do not merely leave users uninformed, they can actively produce mental models that are more conservative than documented industry practice [4], [5]. This places users in a weaker position than simple ignorance would, as a user who believes no data is shared has no reason to seek out controls, read policies, or limit their behaviour in the cabin.

Participants also navigated the vehicle interface using expectations imported from smartphones, searching settings hierarchies for a privacy notice and expressing surprise at its absence. This suggests that the mobile privacy patterns reviewed in our Background are not merely a design resource but the baseline against which users already evaluate vehicle interfaces. Users arrive at the vehicle with a mental model of where privacy information should live, and interfaces that break from this model leave even privacy-literate users, such as our pilot participants, unable to act on their existing protective habits. The full study will probe this explicitly by asking participants to describe where they would expect equivalent information on their phone.

5.2 Implications for vehicle manufacturers

The predominance of interface impossibility outcomes suggests that the transparency gap in older and lower-end vehicles is a content problem rather than a navigation problem. Participants explored the available menus competently and exhausted them. No amount of improved menu structure would have surfaced disclosures that do not exist in the interface. For manufacturers, this implies that transparency cannot be retrofitted through interface reorganisation alone, the disclosures themselves, what is collected, why, and with whom it is shared, need to be present at the point of interaction rather than deferred to printed manuals or external policies that participants stated they would not consult.

The pilot surfaced a social barrier that manufacturer in-

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terface design currently ignores. When framed as a passenger, one participant described disengaging from privacy management entirely, stating they would not adjust settings in another person's vehicle or ask the driver whether they were being tracked. This suggests that even a well-designed privacy interface may go unused by passengers if acting on it requires intervening in the driver's vehicle configuration. A possible design direction is a passenger-facing interface with its own scope of settings, allowing a passenger to view what is collected about them and manage passenger-relevant data flows, such as cabin microphone capture or paired-device data, without altering the driver's configuration. Such separation would respect the social dynamics of shared vehicle use.

Participants' smartphone-derived expectations suggest that manufacturers need not invent vehicle privacy design conventions from scratch. Adopting patterns users already understand from mobile platforms would let manufacturers meet users where their mental models already are, rather than requiring users to learn a new privacy vocabulary for each vehicle brand.

5.3 Protocol refinements

The pilot validated the four-level outcome coding, and in particular the value of distinguishing interface impossibility from participant failure. Six of eleven administered task instances ended in interface impossibility, and treating these as participant failures would have fundamentally mischaracterised the results. Three refinements follow. First, Task 5 was not administered in one session. The full study will fix its administration rules, including whether the passenger framing is presented to all participants or only to passenger-recruited participants. Second, the pilot revealed the need to standardise researcher intervention. Reassurance and hints were given informally, which affects the boundary between partial success and failure. The full protocol will specify permitted probes and the conditions under which participants may be moved to the next task. Third, to capture the absent-disclosure inference systematically rather than relying on spontaneous remarks, the full protocol adds a follow-up after every task coded as interface impossibility, asking participants what they now believe the vehicle does with the data in question and how confident they are in that belief.

5.4 Limitations

The pilot used a 2015 Audi Q5, which predates most modern working definition of a CVs and lacks manufacturer companion-app integration. This confounds a subset of outcomes, for Tasks 3, 4, and 6, we cannot fully distinguish disclosures that are absent from the interface from data practices that are genuinely absent from the vehicle. At the same time, the pilot vehicle retains the core interface functionality of lower-end and older connected vehicles, with settings hierarchies, voice recognition, telematics-adjacent features, which represent a substantial share of the vehicle fleet actually on the road. The pilot therefore speaks to the transparency experience of the many drivers whose vehicles fall below the current state of the art, while motivating the full study's move to a recent-generation vehicle. Full data collection will use a current CV with an integrated data ecosystem and companion application (e.g., a Tesla), enabling all six tasks to target genuinely existing data flows and allowing comparison between interface generations.

Both pilot participants were male, aged 18–29, non-owners, and learned to drive in the same country. Full data collection will recruit for diversity in age, gender, vehicle ownership, and driving background, and will include participants in both driver and front-seat passenger roles.

Pilot participants were recruited through personal networks, a convenience approach appropriate for protocol refinement but not for the full study, which will use broader recruitment channels.

5.5 Future Research Plan

This pilot constitutes the first step of a three-part research plan.

Full think-aloud study. We will conduct the refined protocol with a target sample of N = [20–30] drivers and front-seat passengers using a current-generation CV and its companion application. The full study will characterise the transparency gaps users encounter across interface generations and stakeholder roles, analysed through reflexive thematic analysis of think-aloud and interview data alongside descriptive task-outcome and rating data.

Co-design workshops. Participants from the full study will be invited to co-design workshops in which the documented transparency gaps become design material. Workshop participants will collaboratively develop interface concepts addressing the breakdowns observed in the think-aloud sessions. For example, purpose-specific disclosures, passenger-visible data indicators, and controls that distinguish legally required from discretionary collection. Workshop selection will draw on the concern and behaviour profiles observed in the think-aloud study to ensure diversity of perspectives rather than over-representing highly privacy-motivated users.

Prototype evaluation. The workshop outputs will inform the design of a prototype privacy interface for CVs. We will evaluate this prototype using the same task-based protocol employed in the present study, enabling a direct comparison. The same tasks, attempted on current manufacturer interfaces and on the redesigned interface, with task outcomes, ratings, and think-aloud data as common measures. This comparison closes the loop of the research plan, from documenting transparency gaps, through participatory design, to measuring whether the resulting designs actually improve users' ability to find, understand, and act on privacy information.

6. Conclusion

CVs collect extensive data on drivers and passengers,

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yet the interfaces through which users might understand and manage this collection remain largely unexamined. In this work-in-progress paper, we presented the design of a task-based think-aloud study to evaluate the privacy transparency of CV interfaces, with preliminary observations from pilot sessions. The pilot demonstrated the protocol's analytical yield. Participants could not fully complete any of the six tasks, with the majority of outcomes reflecting information that the interface does not provide rather than information that participants failed to find. Notably, participants interpreted the absence of disclosure as the absence of data collection, suggesting that vehicle interfaces may not only leave users uninformed, but actively foster mental models more conservative than documented industry practice. These observations, alongside the protocol refinements they motivated, position the full study to characterise transparency gaps across a current-generation CV and a diverse participant sample. The findings will subsequently inform co-design workshops and the development of a prototype privacy interface, to be evaluated against current manufacturer interfaces using the same task protocol, completing a research arc from documenting transparency failures to demonstrating whether user-centred design can repair them.

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